

4.2 SUBPHYLUM-CEPHALOCHORDATA

(Gr. *KEPHALE*- HEAD; L. *CHORDA*- CORD)

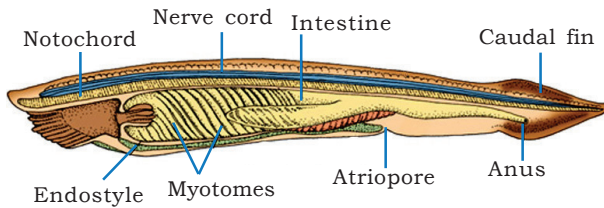


Fig. 4.5 *Branchiostoma*

Cephalochordates are marine animals which lead a burrowing mode of life in shallow sea waters. They are small 'fish like', translucent forms with median fins but without paired fins. They are often described as **typical chordates**, because they possess the principal chordate characters such as **notochord**, **tubular nerve cord** and

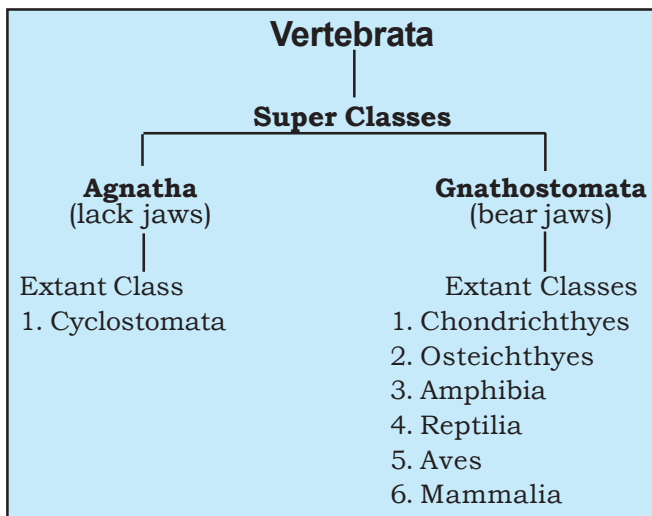
pharyngeal slits throughout their life. The notochord extends from the posterior end to the anterior end of the body beyond the nerve cord. Muscles occur as **myotomes** (blocks of muscles). The coelom is **enterocoelic**. They are **ciliary** or **filter feeders**. The pharynx is surrounded by an ectoderm lined cavity called **atrium**. An **endostyle** is present along the ventral wall of the pharynx. Exchange of respiratory gases occurs mostly across the **external body surface**. Circulatory system is of **closed type** and heart, blood corpuscles and respiratory pigment are absent. Excretion is performed by solenocytes of protonephridiae type. Numerous gonads without gonoducts are present in the pharyngeal region. **Fertilization is external** and development is **indirect** (involves a free swimming larval form).

e.g. *Branchiostoma* (amphioxus or lancelet)

Branchiostoma is often described as a typical chordate. Justify

4.3 SUBPHYLUM-VERTEBRATA/ CRANIATA

Classification of subphylum Vertebrata



The members of the subphylum Vertebrata possess notochord during the embryonic period which is replaced partly or wholly by a cartilaginous or bony vertebral column in the adult. It is true that "**all vertebrates are chordates but all chordates are not vertebrates**". Besides the basic chordate characters, vertebrates have paired appendages which may be 'fins' or 'limbs' and they are with or without jaws. Heart is ventral, muscular with two, three or four chambers. **Mesonephric** or **metanephric** kidneys are the organs of excretion and osmoregulation.